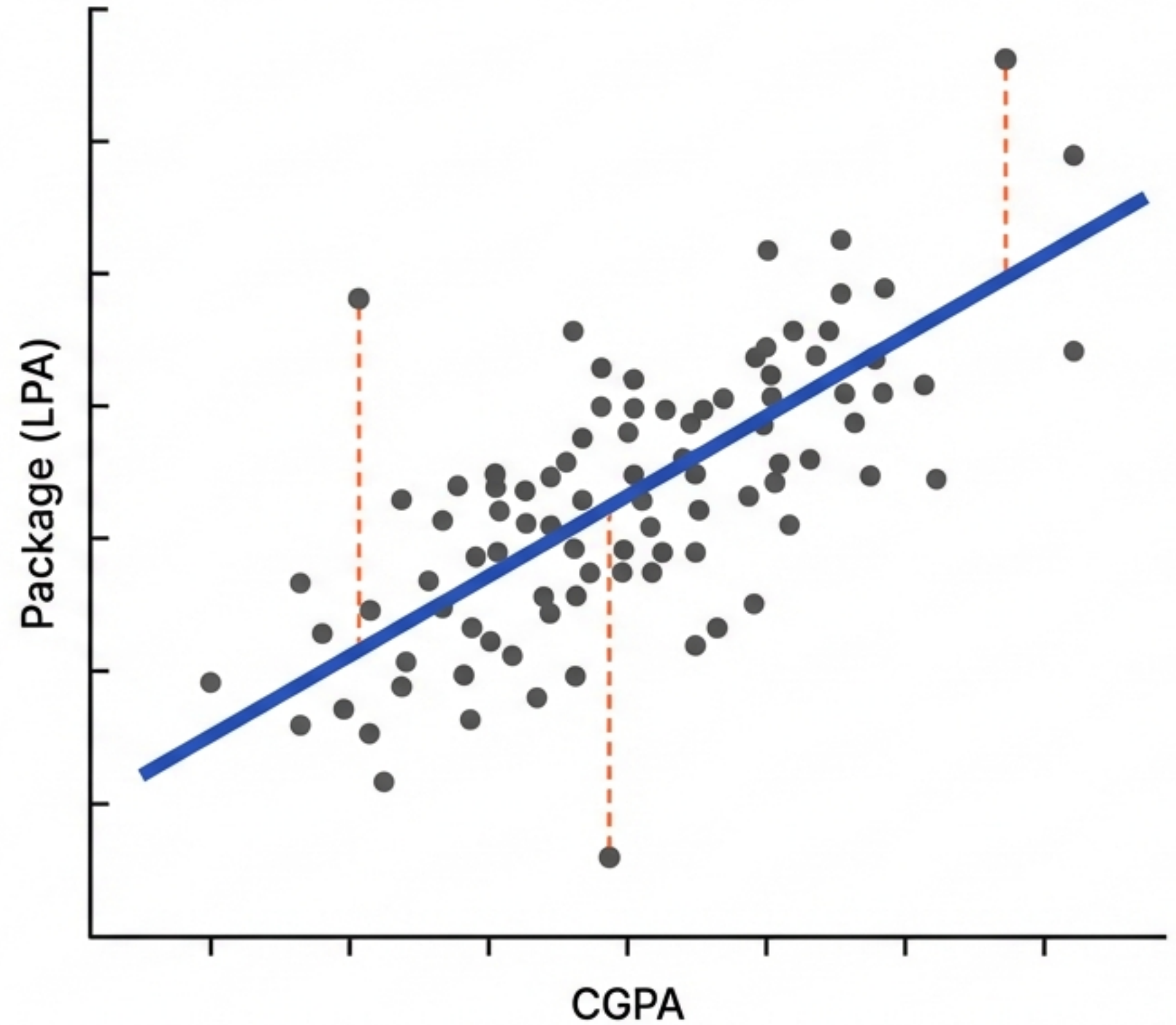


Evaluating Regression Models

From Error to Goodness of Fit

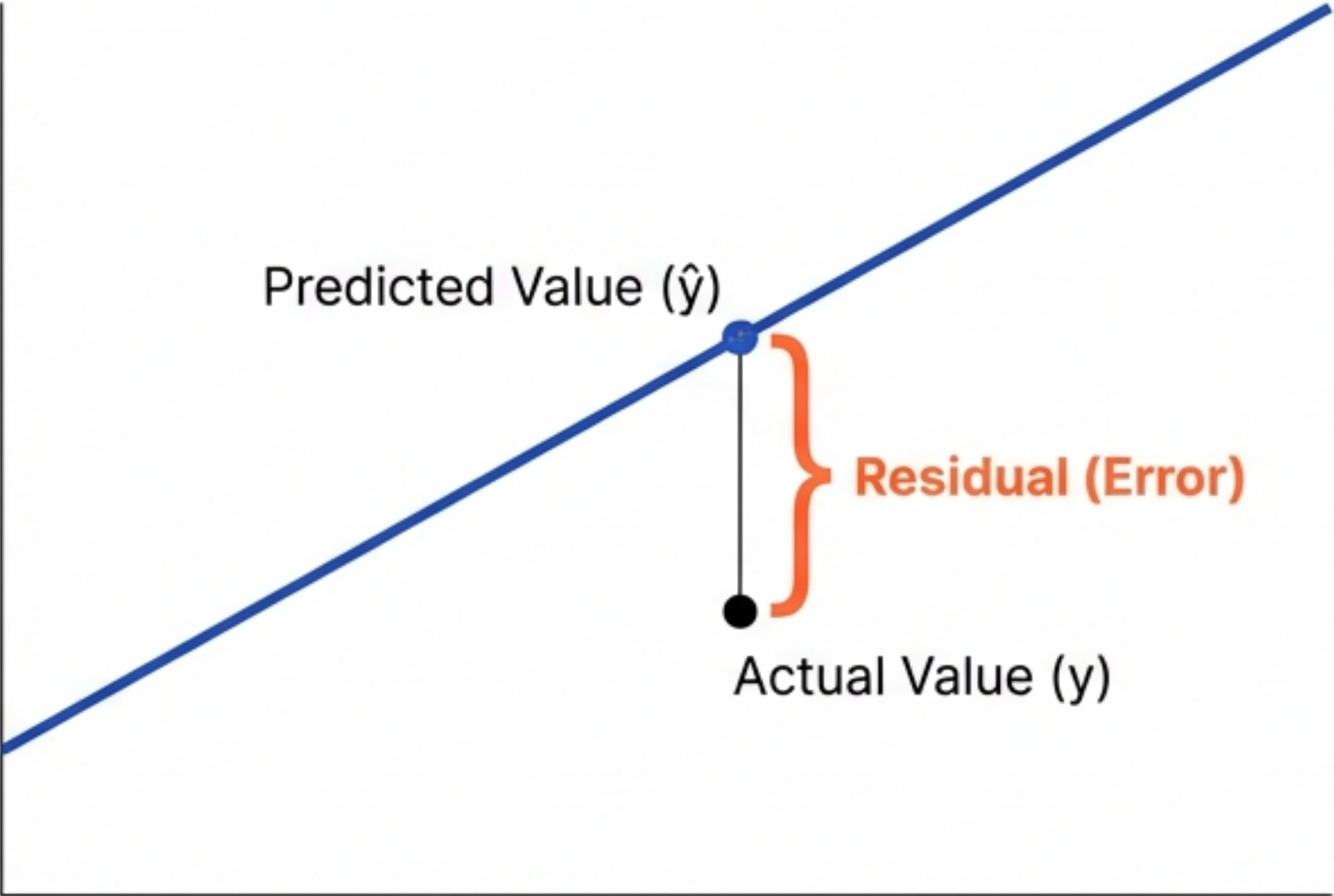
A geometric and mathematical evolution of MAE, MSE, RMSE, and Adjusted R^2 Score.

MODULE: PERFORMANCE METRICS



The Goal: Minimise the Distance Between Prediction and Reality

Student_ID	CGPA (Input)	Package_LPA (Actual)	Predicted_LPA
001	8.5	12.0	11.2
002	7.2	6.5	7.0
003	9.1	15.0	13.8



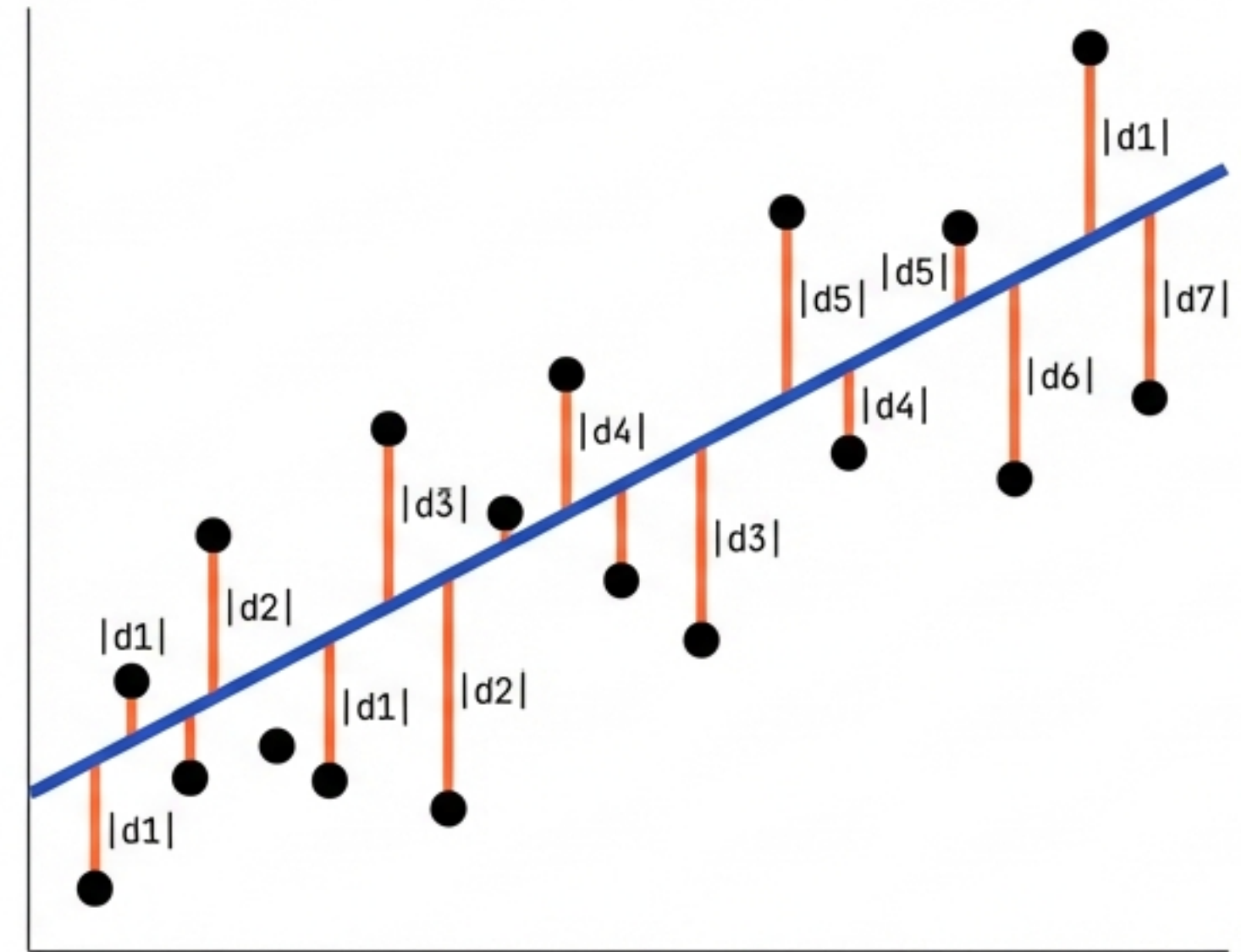
Our objective is to quantify these errors across all 'n' students to determine how much we can trust the model.

Metric 1: Mean Absolute Error (MAE)

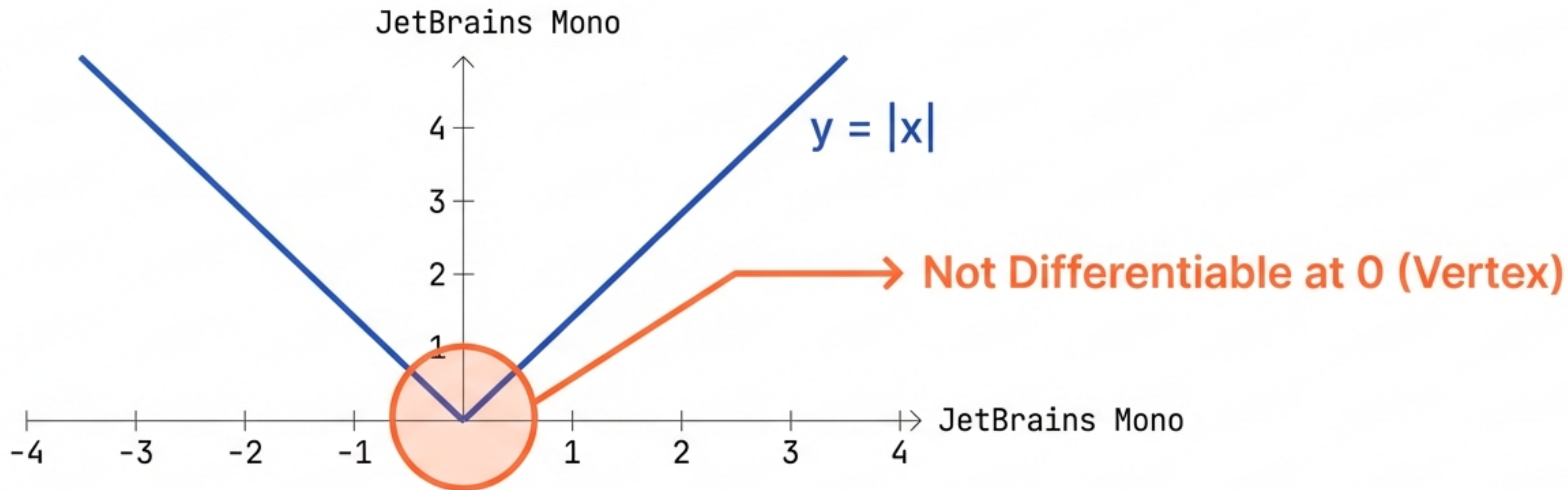
The Intuitive Approach: Summing the Absolute Distances

$$\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

- Calculates the average vertical distance between points and the line.
- Unit Consistency: The error is in the same unit as the target (LPA). An MAE of 1.5 means predictions are off by an average of 1.5 Lakhs.
- Robustness: Treats all errors linearly. Outliers are not disproportionately punished.



The Flaw in MAE: The Math Breaks Down at Zero



Optimisation algorithms like Gradient Descent rely on calculating derivatives (slopes) to find the minimum error. The sharp turn at zero makes the derivative undefined, making convergence computationally expensive and mathematically difficult.

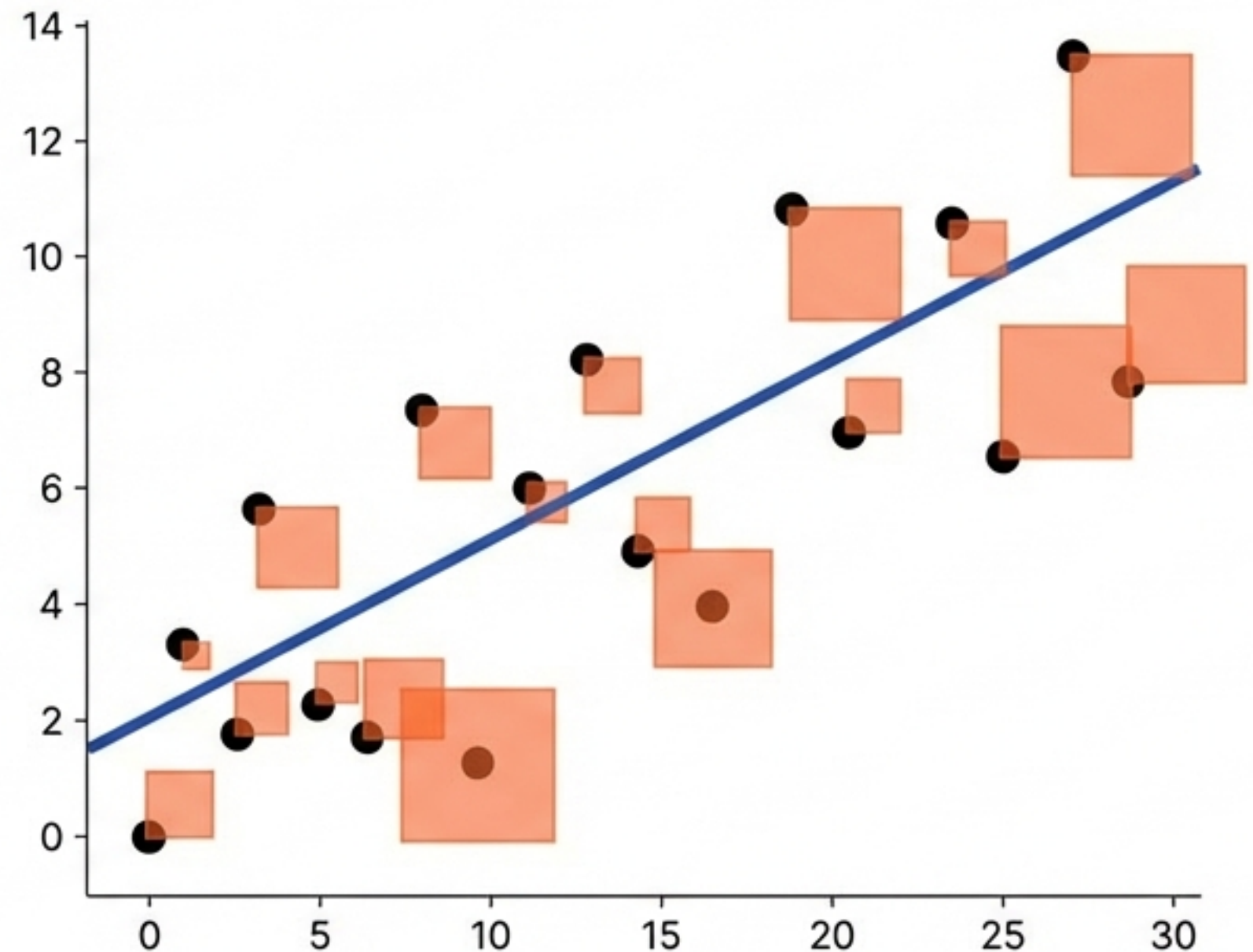
Solution needed: A smooth curve, not a sharp corner.

Metric 2: Mean Squared Error (MSE)

Solving Differentiability by Squaring the Error

$$\text{MSE} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

- By squaring the error terms, we transform the sharp modulus function into a smooth parabola ($y = x^2$).
- Differentiable everywhere, making it the ideal Loss Function for Gradient Descent.
- Geometric Intuition: Minimising the total area of squares attached to the residuals.



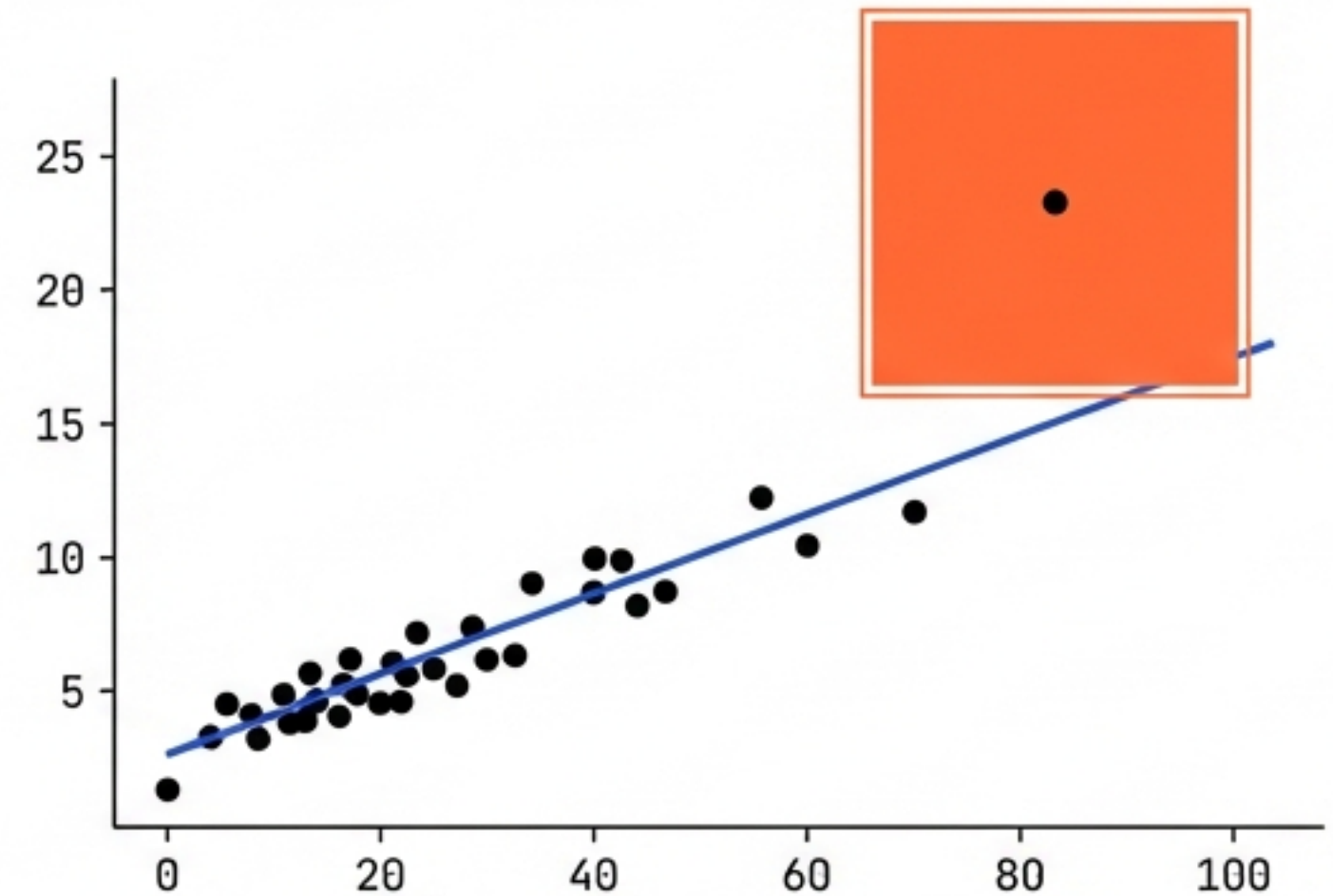
The Double-Edged Sword of MSE

Problem 1: Interpretability



Hard to explain to stakeholders. What is a 'Squared Lakh'?

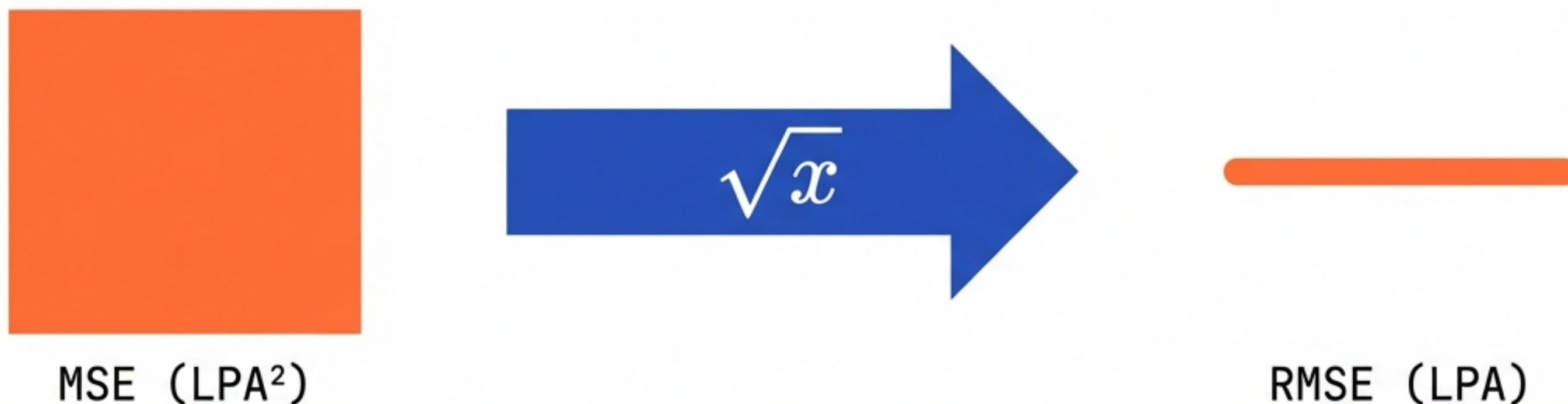
Problem 2: Outlier Sensitivity



Squaring penalises large errors heavily. The model will skew the line drastically to reduce this one massive square, potentially hurting the fit for normal data points.

Metric 3: Root Mean Squared Error (RMSE)

Restoring Interpretability while keeping Differentiability



$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

RMSE returns the error metric to the original unit (LPA), making it human-readable. It is the standard metric for reporting performance in most industries. It solves the unit issue but retains the high penalty for outliers.

Decision Matrix: Choosing the Right Error Metric

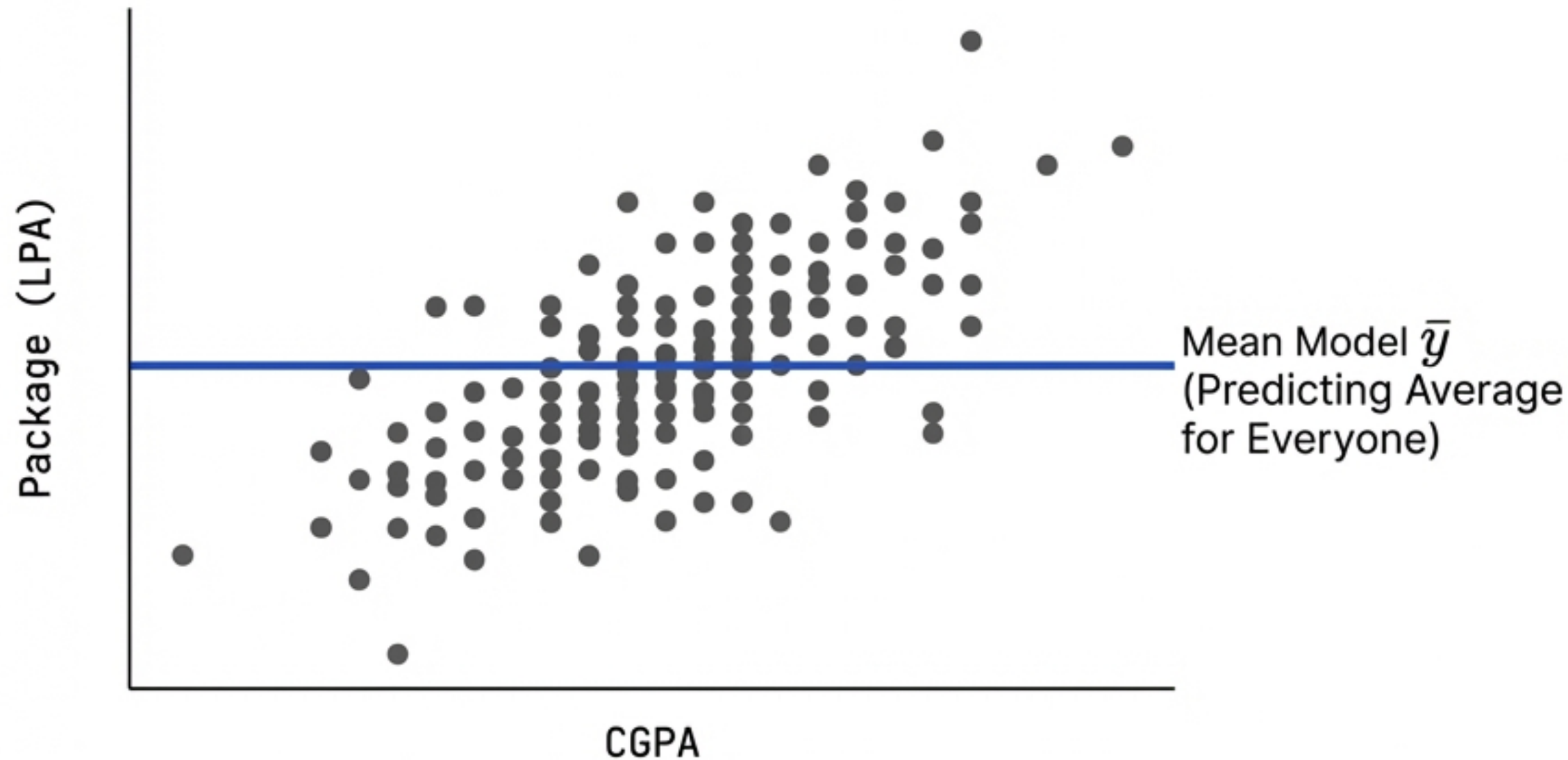
Metric	Primary Attribute	When to Use
MAE	Robust to Outliers	When outliers are present and you do NOT want to penalise them heavily.
MSE	Differentiable	As the internal Loss Function during algorithm training.
RMSE	Interpretable Units	For reporting final model performance to clients/stakeholders.

Insight: Always calculate all three. If RMSE is significantly higher than MAE, it confirms the presence of large **outliers** in your data.

Is an Error of 1.5 LPA Good? It Depends.

We need a baseline to compare against.

The Worst Case Scenario: The Mean Line



Imagine a naive model that ignores the input (CGPA) and just predicts the average package for every student. This is our baseline.

Our regression model is only useful if it performs **better** than this line.

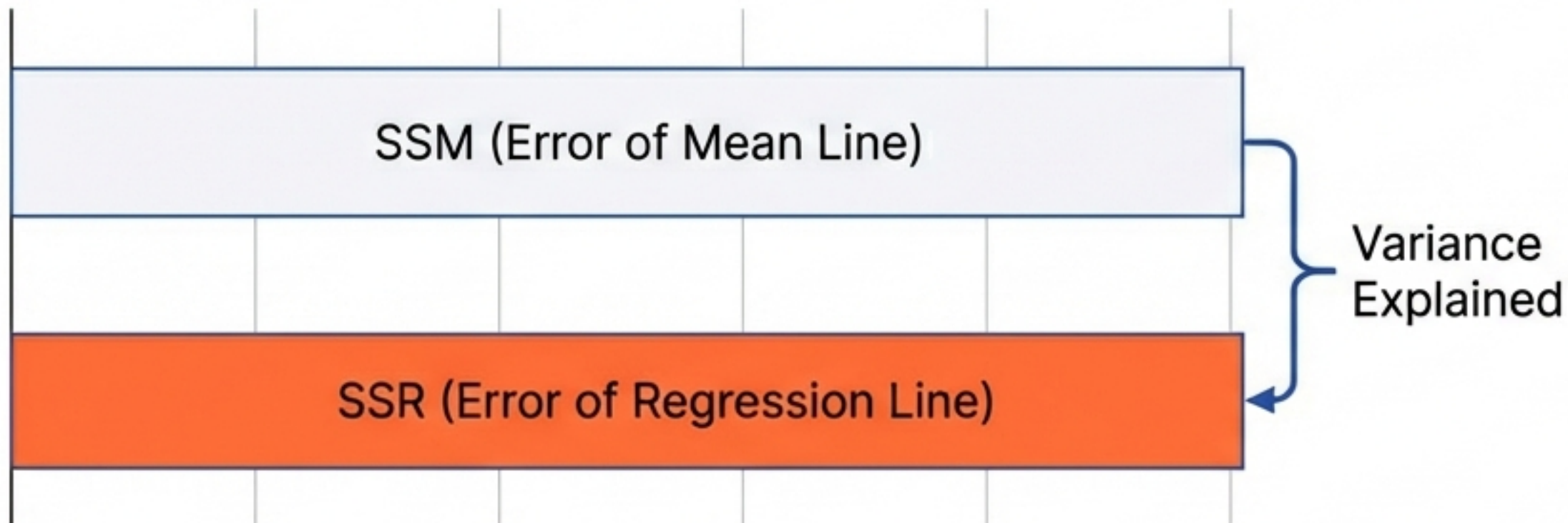
Note a. (not requested, so omitted)

Metric 4: The R^2 Score (Coefficient of Determination)

$$R^2 = 1 - \frac{SSR}{SSM}$$

SSR = Sum of Squared Errors of Regression Line

SSM = Sum of Squared Errors of Mean Line

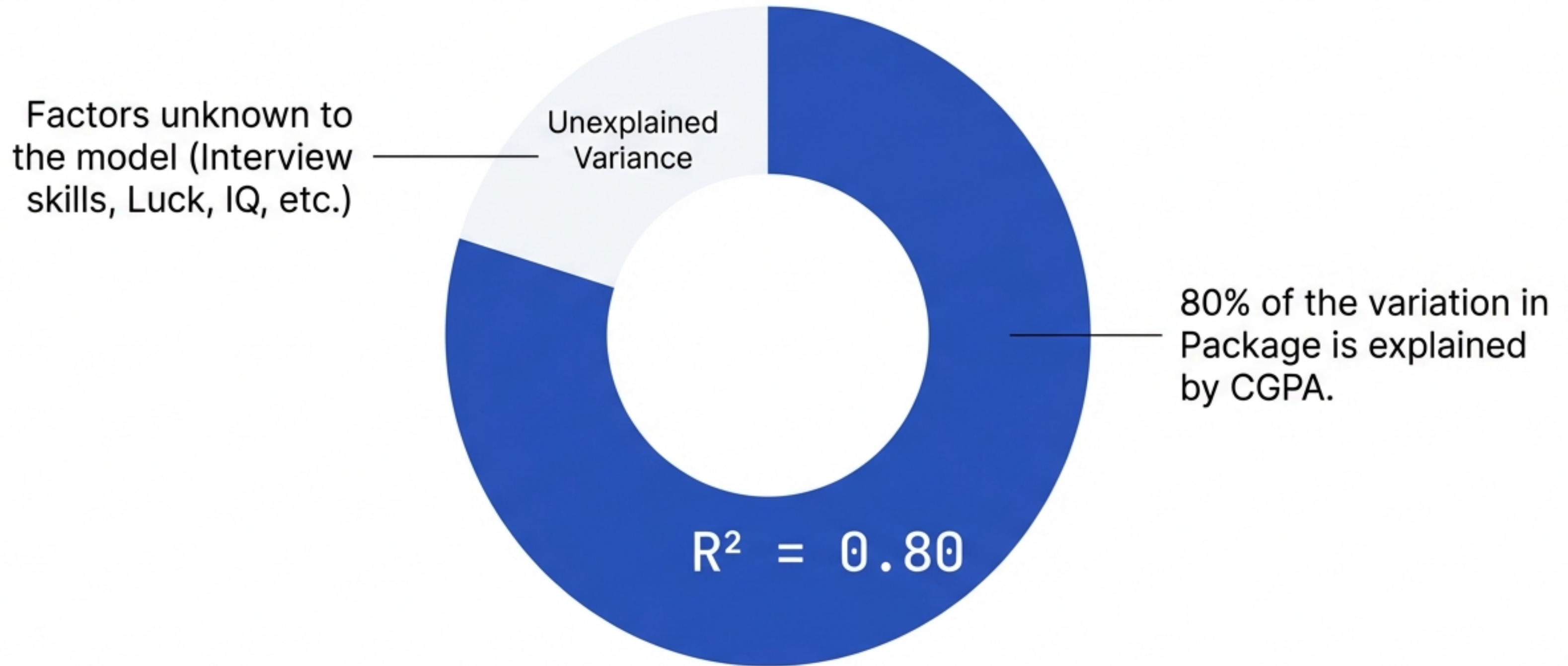


Measures “Goodness of Fit”

It compares our model's error against the mean model's error.

- $R^2 = 1$: Perfect Fit
- $R^2 = 0$: No better than the mean
- $R^2 < 0$: Worse than the mean (**broken model**)

Decoding R^2 : The CGPA Example



An R^2 of 0.80 gives us confidence that CGPA is a strong predictor, but not the only factor.

The Flaw in R^2 : It Can Be Tricked

Step 1

CGPA	Package
8.5	12.5 LPA
7.8	10.2 LPA
9.1	14.0 LPA
6.5	8.1 LPA
8.0	11.5 LPA

$$R^2 = 0.80$$

Step 2

CGPA	Temperature	Package
8.5	28°C	12.5 LPA
7.8	32°C	10.2 LPA
9.1	25°C	14.0 LPA
6.5	30°C	8.1 LPA
8.0	27°C	11.5 LPA

$$R^2 = 0.801$$

Problem: R^2 never decreases when you add new features, even if they are useless (like 'Temperature on Interview Day'). This allows misleading inflation of the score by adding junk data.

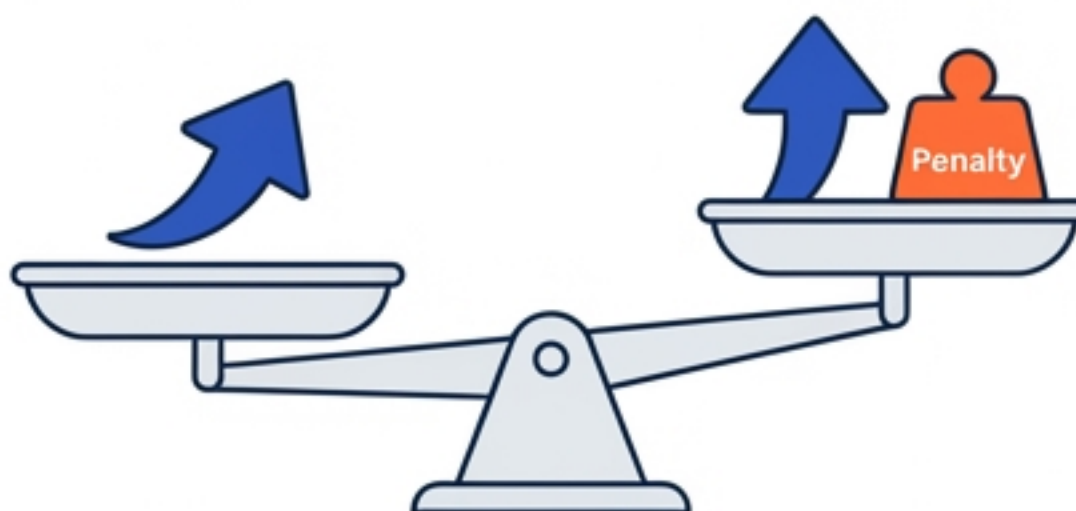
Metric 5: Adjusted R² Penalises Useless Complexity

$$R^2_{adj} = 1 - (1 - R^2) \frac{n - 1}{n - 1 - k}$$

n = Number of rows (students)

k = Number of predictors (columns)

Model Improvement
(Higher R²)



Complexity Penalty
(Higher k)

The Mechanism: As 'k' increases, the penalty term grows. If the new feature doesn't improve the model enough to offset the penalty, the Adjusted R² score will drop.

The Evidence: R^2 vs. Adjusted R^2

Model A: CGPA Only

R^2 : 0.77

Adjusted R^2 : 0.76

Model B: CGPA + Random Numbers

R^2 : 0.78 (Increased - False Positive)

Adjusted R^2 : 0.75 (Decreased - True Negative)

The Adjusted R^2 correctly identified that the Random Number column was junk, penalising the score despite the raw R^2 increasing.

Summary: What Should You Report?



Internal Analysis: Calculate MAE, MSE, and RMSE.

Look for gaps between MAE and RMSE to detect outliers.



Client Communication: Report RMSE and Adjusted R^2 .

Clients understand 'Average Error in Lakhs' and 'Percentage of Variance Explained'.



Context is King.

An error of 1.5 is excellent for salary prediction but catastrophic for precision engineering. Always compare against domain standards.